

# ◆ Advanced Subnetting & Masking Techniques (IPv4)

## 1 AD Rule 2.0 – Fast Subnet Identification Method

AD Rule 2.0 is a shortcut to quickly determine:

- Subnet ID
- Directed Broadcast Address (DBA)
- Specific Nth subnet

Without calculating full IP ranges manually.

### ◆ Core Principle

To find the **Nth subnet**:

[  
    \textbf{Convert (N – 1) to binary}  
]

Why?

Because subnet numbering starts from **0** in binary.

| Subnet Number | Binary Representation |
|---------------|-----------------------|
|---------------|-----------------------|

|      |             |
|------|-------------|
| 1st  | 0           |
| 2nd  | 1           |
| 8th  | 7 (000111)  |
| 17th | 16 (010000) |

### ◆ Step-by-Step Procedure

1. Identify default Network ID (based on class)

2. Determine how many bits are borrowed
3. Convert  $(N - 1)$  into binary using borrowed bits
4. For Subnet ID → set remaining host bits to 0
5. For DBA → set remaining host bits to 1

## 2 Subnet Mask Mechanics

A subnet mask tells us:

- Which bits are Network + Subnet (1s)
- Which bits are Host (0s)

### Rule Summary

| Mask Bit | Meaning                |
|----------|------------------------|
| 1        | Network ID + Subnet ID |
| 0        | Host ID                |

### Default Classful Masks

| Classes | Default Mask  | NID Bits | HID Bits |
|---------|---------------|----------|----------|
| A       | 255.0.0.0     | 8        | 24       |
| B       | 255.255.0.0   | 16       | 16       |
| C       | 255.255.255.0 | 24       | 8        |

### Host Capacity Formula

If  $n$  = number of Host ID bits:

- Total IPs per subnet =  $2^n$
- Usable hosts =  $2^n - 2$

(Subtracting Subnet ID & Directed Broadcast Address)

### 3 Case Study – Class B Example

Given:

Network: 157.157.0.0

Class B → Default 16 network bits

Required: 64 subnets

Since:

[  
 $2^6 = 64$   
]

👉 Borrow 6 bits from Host ID  
Remaining Host bits =  $16 - 6 = 10$

#### Calculated Subnets Using AD Rule 2.0

| Target | (N-1) | Binary | Subnet ID     | DBA             |
|--------|-------|--------|---------------|-----------------|
| 8th    | 7     | 000111 | 157.157.28.0  | 157.157.31.255  |
| 17th   | 16    | 010000 | 157.157.64.0  | 157.157.67.255  |
| 28th   | 27    | 011011 | 157.157.108.0 | 157.157.111.255 |
| 61st   | 60    | 111100 | 157.157.240.0 | 157.157.243.255 |

Each subnet contains:

[  
 $2^{10} - 2 = 1022$  usable hosts  
]

### 4 Theoretical vs Practical Bit Borrowing

#### Practical Networking Rule

- ✓ Bits must be borrowed from the **most significant side (left side)** of the host portion.
- ✓ Subnets must produce **sequential IP ranges**.

## Theoretical (Exam-Oriented) Borrowing

Some competitive exams (e.g., ISRO-type problems) may allow:

- Borrowing bits from non-MSB positions
- Non-contiguous subnet masks

This is mathematically valid but:

- ✗ Not used in real networking
- ✗ Produces non-sequential address blocks

Real-world routers require **contiguous masks**.

## 5 Non-Standard Mask Example

Mask: 255.255.255.44

Binary of 44:

00101100

1s are at positions:

- 32
- 8
- 4

Total subnet bits = 3

Number of subnets =  $2^3 = 8$

Even though bits are non-contiguous, mathematically it defines subnets.

⚠ But again — not valid in practical routing environments.

## **6 Key Concept Reinforcement**

- Subnet mask = tells how many bits are borrowed
- Subnet ID = remaining host bits all 0
- DBA = remaining host bits all 1
- Usable hosts =  $2^n - 2$
- AD Rule 2.0 = Convert  $(N - 1)$  to binary

## **7 Practical Networking Perspective**

Modern networking uses:

- CIDR (Classless Inter-Domain Routing)
- Contiguous subnet masks
- VLSM (Variable Length Subnet Masking)

Classful subnetting is mainly used for:

- Academic exams
- GATE preparation
- Competitive tests